



Don't Be A Loser!

You don't know
how much you
don't know.

Read

electronics
FOR YOU

to keep up-to-date
with the tech world



To subscribe,
visit efy.in or
contact us at: 011-4059660
Email: support@efy.in

OR

Scan
This
Code



Also available at all leading magazine outlets

battery-operated devices remains challenging. This has driven the development of low-power SoCs and heterogeneous microcontrollers, such as RISC-V combined with ARM cores, with companies like TI, Microchip, and Renesas leading the way.

Connectivity advances. 5G with NB-IoT provides low-bandwidth continuous connectivity ideal for IoT devices, while the SG20 forum within the ITU defines IoT-specific protocols that will extend into 6G.

Open source innovation. Platforms like RISC-V allow developers to customise cores, similar to i3, i5, and i7 versions, adding or removing features as needed. These cores are available in Verilog, giving engineers the flexibility to design custom semiconductors.

AI is also being applied in the healthcare sector, drone technology, and predictive maintenance. In energy management, AI is continuously monitoring electricity usage and sensor information, identifying anomalies and locating equipment for preventive action. In smart energy, usage is monitored, warnings are issued, and automatic disconnection is possible when the threshold is reached, eliminating the need for manual readings while presenting both opportunities and challenges.

In India, the smart meter market exceeds 200 million units, presenting a huge opportunity. The challenge is that the buyer is not the user, and the user may have little incentive to operate the device correctly, sometimes even attempting to tamper with it. The issue of smart meter security is addressed by AI-enabled smart meters, which analyse sensor data to detect anomalies or tampering attempts.

Beyond energy management, AI is also changing wearable technology and motion control. In

the medical field, robotic surgery provides accuracy that is sometimes beyond human reach. In manufacturing, AI is used in motion control. Autonomous vehicles are also being developed worldwide. Although there are challenges in the adoption of autonomous vehicles in India, it is likely to become a reality in the near future.

AI is also revolutionising electronics and hardware design. From circuit design and PCB layout to verification, validation, and system-level optimisation, multiple stages can now be AI-driven.

Circuit design includes automated generation, component selection, and bill of materials (BoM) creation. PCB design incorporates intelligent placement, routing, thermal management, and power and ground plane optimisation.

Verification and validation leverage simulation, predictive analysis, and synthetic data generation to ensure performance and reliability. System-level optimisation enables architectural exploration and material innovation, such as the use of silicon carbide components where advantageous.

The road ahead

An emerging approach involves embedding lightweight AI directly within systems. This enables real-time feedback to both the generative engine and the system itself, allowing continuous refinement and adaptive learning. Although still evolving, low-power TinyML solutions and cloud-connected architectures are being explored to aggregate data from multiple embedded systems, enabling continuous improvement and increasingly autonomous operation. **EFY**

The article is based on a presentation last year at EFY Expo in Pune titled 'Harnessing AI to Consolidate and Scale Embedded Systems & IoT Innovations,' by Tapan Kapoor, Co-Founder, Avicomm Solutions Pvt Ltd. It has been transcribed and curated by Saba Aafreen, Technology Journalist at EFY.